ESI Assignment – Quick Guide

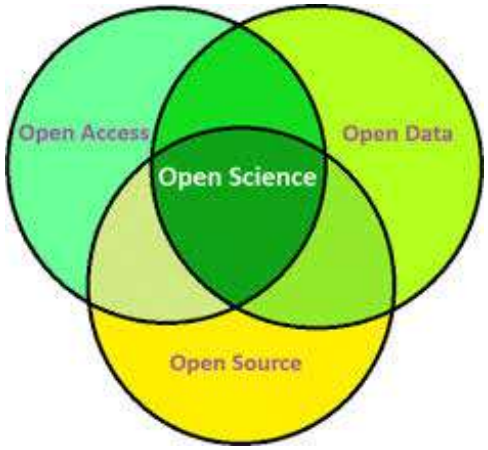
18 March, 2026

Assignment goals (1/2)

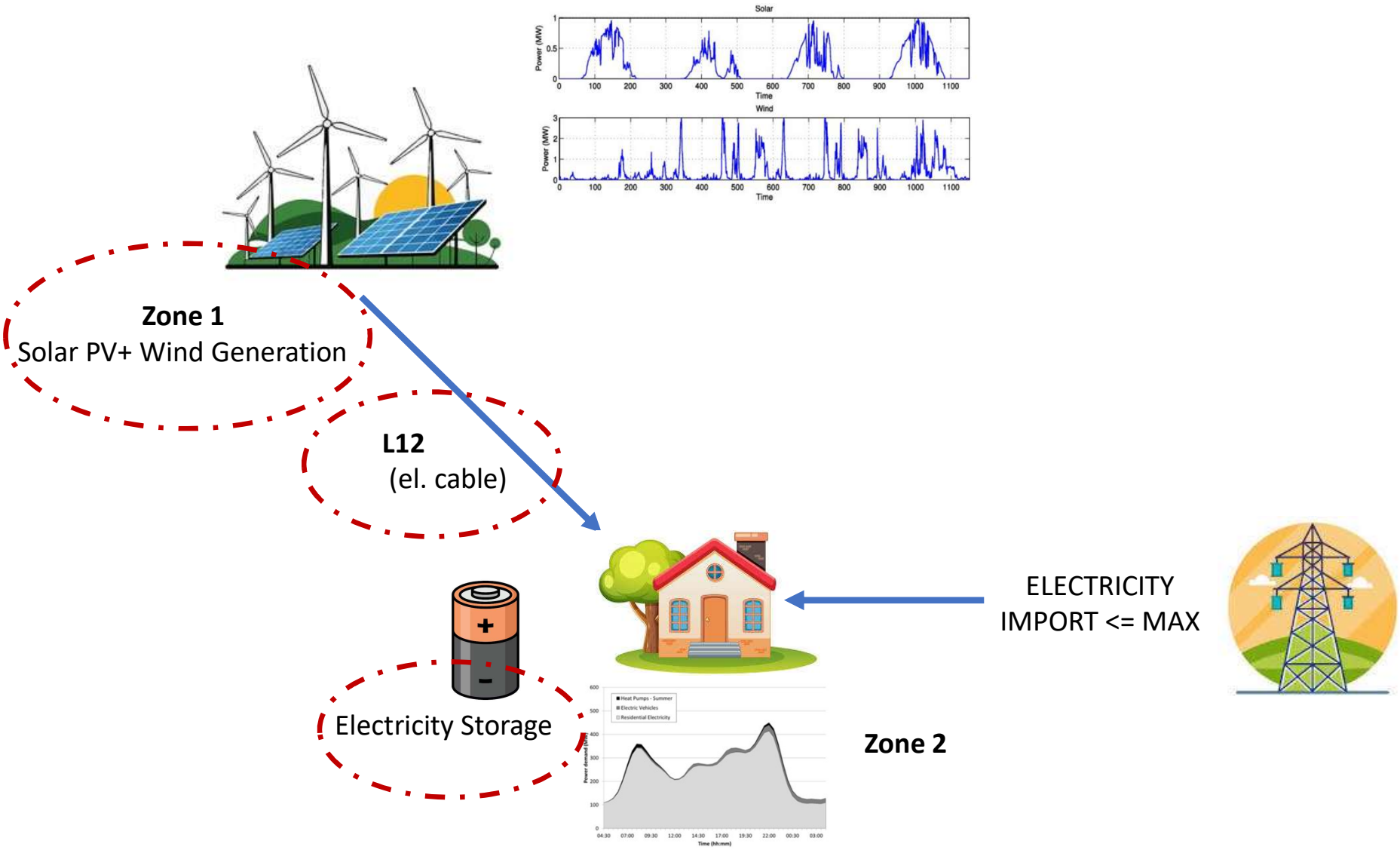
- Apply the theoretical knowledge we learnt in the ESI course to a (simple) three-zone energy transition problem
- Learn to combine various energy conversion, transport and storage technologies into a coherent system design
- Understand the trade-offs between RES integration, reliability of supply and investment costs

Assignment goals (2/2)

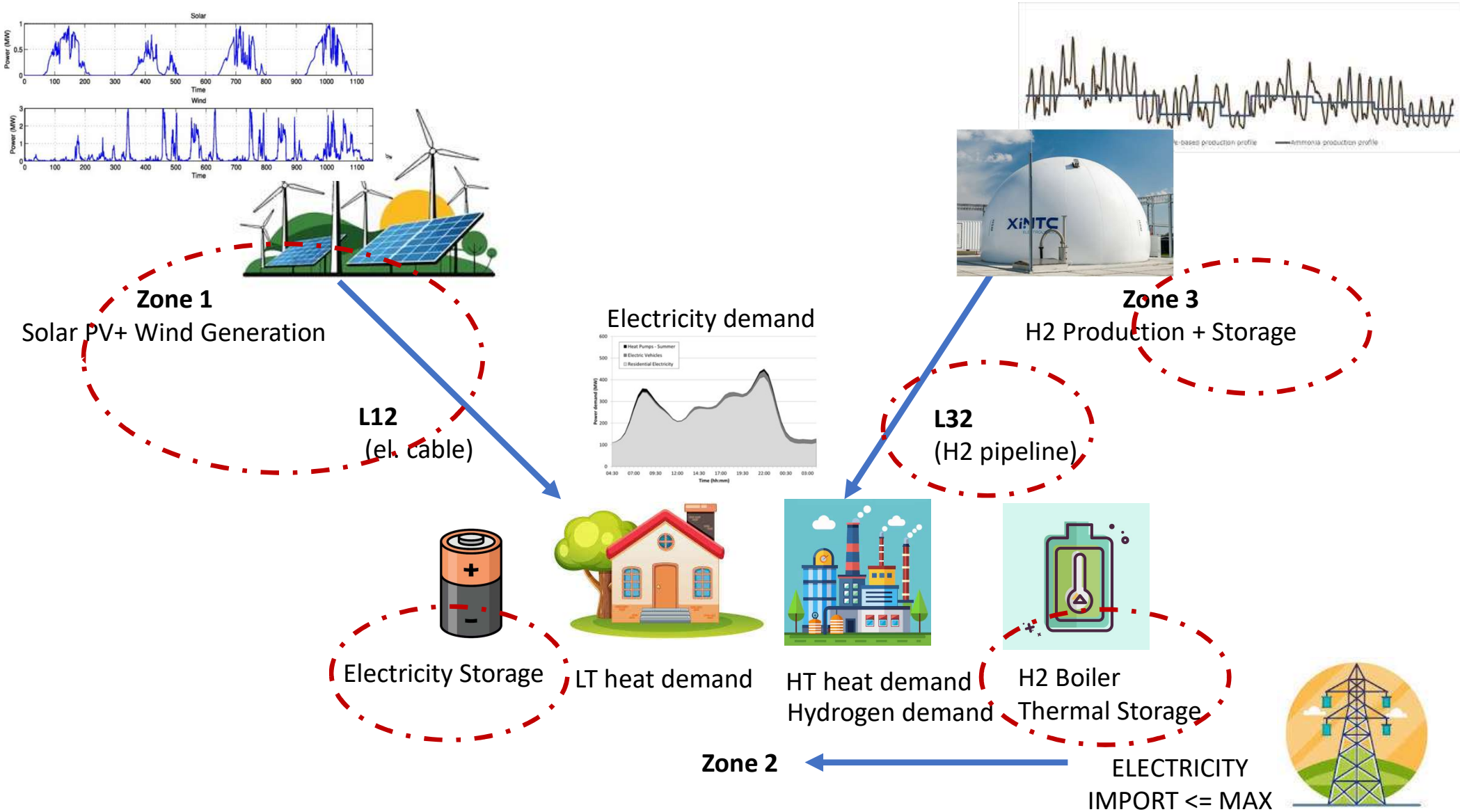
- Explore locational effects and operational strategies for storage
- Explore the utilization of electricity and hydrogen infrastructure
- Reflect on design limitations
- Practice open science by combining data, coding, analysis and reporting in one tidy package



Part 1: Renewable Electricity


Solar
Power (MW)
Time
Wind
Power (MW)
Time
Zone 1
Solar PV+ Wind Generation
L12
(el. cable)
Electricity Storage
Zone 2
ELECTRICITY
IMPORT <= MAX

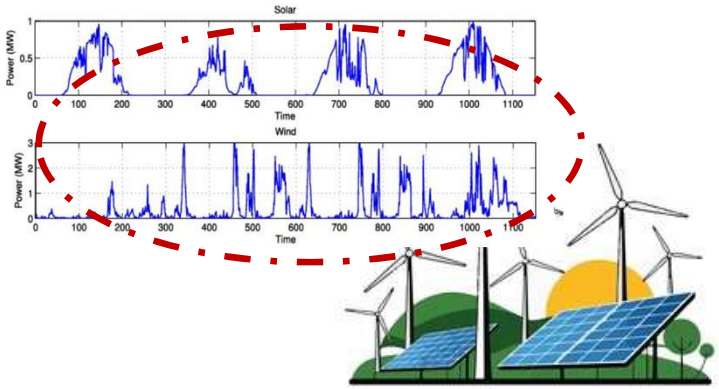
Part 2: Decarbonization of heat and industry, add H2


Solar
Power (MW)
Time
Wind
Power (MW)
Time
Zone 1
Solar PV+ Wind Generation
L12
(el. cable)
Electricity demand
Heat Pumps - Summer
Electric Vehicles
Residential Electricity
Power demand (MW)
Time (hh:mm)
L32
(H2 pipeline)
Zone 3
H2 Production + Storage
XiNTC
Electricity Storage
LT heat demand
HT heat demand
Hydrogen demand
H2 Boiler
Thermal Storage
Zone 2
ELECTRICITY
IMPORT <= MAX

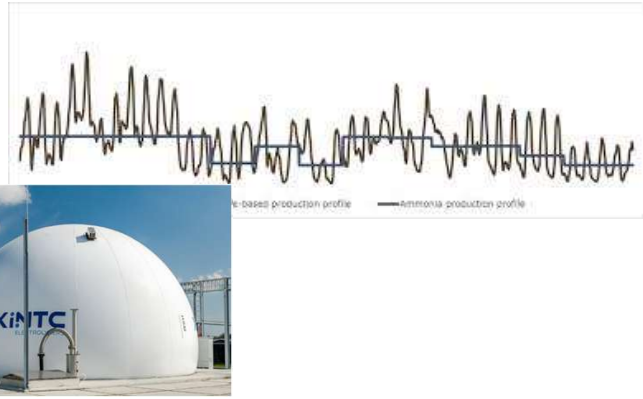
KPIs for comparing designs

- Amount of RES energy curtailment
- Amount of unserved electricity, heat or hydrogen demand and associated monetary penalties
- Total conversion and transport losses (excluding RES)
- Total annuitized system cost (CAPEX, OPEX, IMPORT, PENALTY)
- Utilization factor of the L12 transmission line
- Number of hours the L32 pipeline is at its maximum capacity
- Number of hours that the H_2 reservoir is empty or full
- Any other ideas?

Part 2: Reflection on design limitations – QUALITATIVE only



Zone 1
Solar PV+ Wind Generation

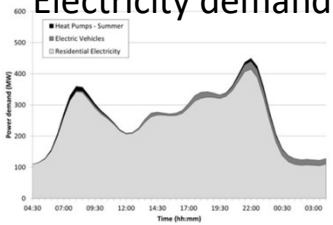


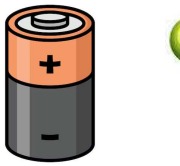
Zone 3
H2 Production + Storage

L12
(el. cable)

L32
(H2 pipeline)

Electricity demand

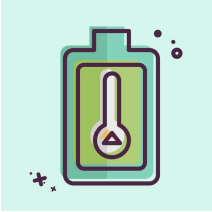




Electricity Storage

LT heat demand

HT heat demand
Hydrogen demand

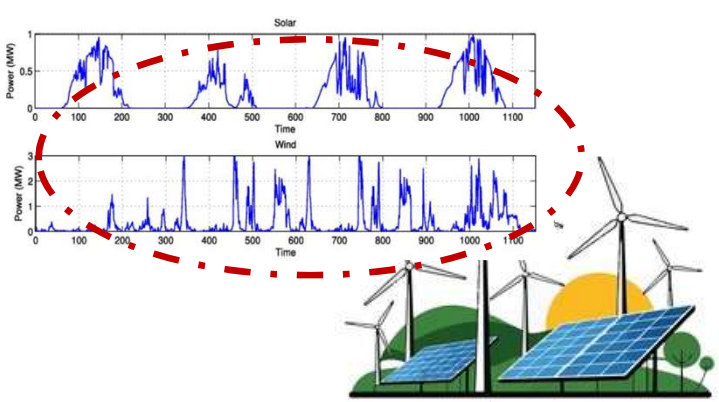


H2 Boiler
Thermal Storage

Additional technologies?

Zone 2

ELECTRICITY
IMPORT <= MAX



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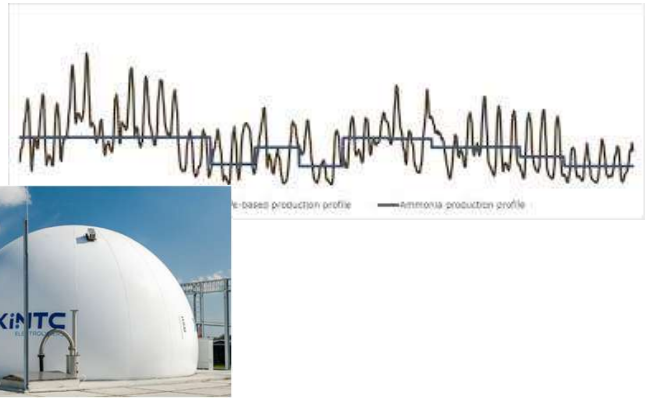
Solar PV+ Wind Generation

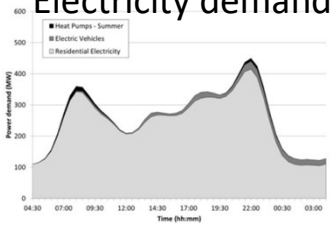


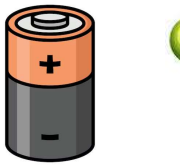
Zone 3
H2 Production + Storage

L12
(el. cable)

L32
(H2 pipeline)

Electricity demand





Electricity Storage

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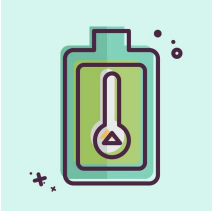

H2 Boiler
Thermal Storage

Additional technologies?

Zone 2

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Part 3 (Bonus): Robustness under Uncertainty - QUANTITATIVE

- Weather uncertainty
- N-1 analysis

Assignment Logistics

- Work in groups of 2 students
- Make sure to use the unique input parameters corresponding to your group number
- Start working in the provided notebooks, they already have functions for reading the data and creating capacity factor time series for wind and solar
- Finish Part 1 this week and begin on Part 2 (design)
- Finish Part 2 next week (design and operation) + Reflection on limitations
- Groups who are quick may have time for Part 3 (bonus)
- Use markdown cells to explain your assumptions, design criteria and operational strategies, explain trade-offs, analyze your results etc.
- Support your results with attractive visualizations.
- Upload your Jupyter notebooks, AI and Collaboration statements to Brightspace by Monday, Mar. 30th at 23:59
- Presentations to be planned for Mar. 31st and Apr. 2. Good luck!